PROJECT ADDENDUM

Staff Attendance & Multi-Model Accuracy

An attendance module and a practical model orchestration plan for the hypermarket analytics SaaS.

Core design

Use staff credentials for authoritative clock events. Use computer vision as a separate, optional operational signal. Combine independent models, temporal checks, and business rules for analytics estimates; send uncertain cases to review.

Supplement to the Hypermarket Camera Analytics project brief

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1. Attendance module for the SaaS

Add attendance as a distinct module inside the shared SaaS product. It can reuse organization, store, department, user, and role management while keeping attendance records separated from customer analytics, inventory data, and restricted security clips.

Recommended source of truth: a staff member clocks in through an authenticated web/mobile flow, store tablet, PIN, or badge. A camera cannot reliably establish employee identity or prove paid working time by itself.

Attendance capabilities

Capability	Details
Clock in/out	Web/mobile, shared store tablet, PIN, badge/NFC, or approved HR terminal. Record source, store, local time zone, and server timestamp.
Breaks	Start/end break, paid/unpaid type if policy requires, missed-break exception.
Shifts	Published schedules, shift swaps/approvals, grace periods, overtime flags.
Timesheets	Daily/weekly hours, exception queue, employee review, manager approval, export.
Corrections	Missed punch request, reason, manager decision, before/after audit trail.
Integrations	Payroll/HR export or API; employee roster, departments, leave, and schedules.

Optional camera signal

A camera may estimate anonymous occupancy in a staff-only work zone or confirm that a designated area is active during a shift. Treat it as an operational occupancy signal, not an employee identity match and not a clock event. It must not automatically mark someone late, absent, on break, or off-task.

2. Attendance workflow and data

Step	Workflow
1. Schedule	Manager publishes shifts and assigns employees/store/department.
2. Authenticate	Employee signs in with account plus PIN/badge or approved device authentication.
3. Clock event	System records clock-in/out or break event, source device, timestamp, and store.
4. Validate	Rules flag missing punches, overlaps, early/late punches, or unapproved location. Flags are exceptions, not automatic discipline.
5. Review	Employee reviews own time; manager resolves exceptions and records reason.
6. Export	Approved timesheet exports to payroll/HR; export result and period are logged.

Suggested attendance data fields

Record	Suggested fields
Employee	employee_id, organization_id, store_id, department_id, role, status, external_hr_id
Shift	shift_id, employee_id, planned_start/end, time_zone, department, approval_status
Time event	event_id, employee_id, event_type, server_timestamp, device_id, source_method, store_id
Exception	exception_id, event_id, reason_code, employee_note, manager_decision, reviewer, audit timestamps
Timesheet	period, planned_minutes, recorded_minutes, approved_minutes, break totals, payroll export status
Device	device_id, store_id, type, attestation/status, last_seen, allowed methods

Important implementation details

- Use server time as the canonical event time; retain local timezone for display and payroll rules.
- Support offline store tablet mode only with signed local events and later synchronization, if required.
- Prevent duplicate punches with idempotency keys and clear user feedback.
- Make edits append-only corrections with a reason and audit trail; do not silently overwrite the original punch.
- Give employees a way to review their records and raise a correction request.

3. Multi-model design for better accuracy

Use a series pipeline for sequential checks and an ensemble where independent models can review the same image. This can improve robustness, but it does not guarantee a particular accuracy level. Measure each task in the actual store conditions and expose uncertainty instead of forcing a confident answer.

Stage	Models/checks	Purpose
A. Input quality gate	Blur/exposure/occlusion checks; camera health.	Reject unusable frames early; produce UNKNOWN or VIEW BLOCKED.
B. Parallel perception	Two independently trained person detectors; a separate produce/bin detector or segmenter.	Compare detections and reduce dependence on one model.
C. Fusion	Match overlapping detections by location/overlap; combine calibrated confidence; retain disagreement.	Merge agreement; flag disagreement for lower confidence/review.
D. Temporal tracking	Multi-object tracker plus trajectory/zone rules over time.	Stabilize counts and dwell; remove one-frame noise.
E. Task-specific validation	Bin fill estimator; time persistence; POS/inventory reconciliation; camera-region rules.	Convert image evidence to product availability or operational event.
F. Decision policy	Thresholds, confidence bands, persistence, human confirmation.	Alert only when evidence meets a measured policy.

Example serial flow

Frame quality -> parallel people/product detections -> detector agreement/fusion -> temporal tracker -> zone/bin logic -> sales/inventory context -> confidence and alert rule -> dashboard or staff review.

Ensemble methods

- Detection ensemble: run two diverse detectors on sampled frames or critical conditions; reconcile boxes using overlap and class agreement. Do not count duplicate boxes twice.
- Temporal ensemble: combine evidence over multiple frames so one missed or false detection does not immediately change status.
- Sensor/data fusion: compare vision fill estimates with POS sales, receiving, restock confirmations, and scale readings where available. Keep the source of each signal visible.
- Rule ensemble: apply store-defined zone, threshold, time-persistence, and data-quality checks after model inference.

4. Attendance + camera boundary

Need	Recommended source	Camera role
Who clocked in?	Authenticated employee account plus PIN/badge/device method.	None for identity. Do not infer identity from appearance.
When did the punch happen?	Server timestamp and signed device event.	May flag a camera/device outage, not replace the timestamp.
Is a work zone active?	Optional anonymous occupancy estimate.	Counts people in the zone; does not name them or decide attendance.
Was the punch in an allowed store?	Approved kiosk/device location or store terminal.	Optional broad context only; never sole proof.
Are hours correct?	Schedule + punch events + employee/manager review.	No camera-based automatic time deduction.

Prohibited or high-risk shortcuts

- Do not link face templates or persistent camera tracks to employee records for attendance.
- Do not infer absence, tardiness, breaks, productivity, or misconduct from camera analytics alone.
- Do not silently repurpose cameras installed for security or customer analytics into worker monitoring.
- Do not use a model score as payroll truth; provide exception review and correction.

Access and tenant separation

For a multi-tenant SaaS, each organization is a tenant boundary. Enforce tenant_id in every attendance query and authorization decision. Store managers see only assigned stores; employees see their own records; payroll administrators see approved exports; system support access is time-limited and audited.

5. Accuracy plan and evaluation

Set quality gates per feature. Do not advertise one overall "high accuracy" number: traffic counts, bin fill, person tracking, and security cues are different tasks with different costs of error. A false security alert and a small fill-estimation error are not equivalent.

Feature	Evaluation measure	Operational gate
Attendance punches	Duplicate/missing event rate; sync delay; correction rate.	Credential event and audit logic validated; payroll totals reconcile on pilot periods.
Traffic count	Absolute count error and percentage error vs manual sampled counts.	Set tolerance by store/zone and volume; report coverage/unknown periods.
Dwell	Error distribution on manually reviewed sample visits.	Check track fragmentation and crowded periods separately.
Produce availability	Precision/recall for LOW/EMPTY; fill estimate error; blocked/unknown rate.	Tune per bin/product; require persistent state and staff checks.
Security review	False alerts per camera/day, reviewer usefulness, missed-event audit sample.	Shadow mode first; only limited queue after separate approval.

How to validate the model ensemble

- Create a baseline with one model for each task; record performance, latency, and cost.
- Add a second diverse model and compare on a held-out set from different times/cameras. Keep it only if it improves the target metric without unacceptable latency or false alerts.
- Calibrate confidence on validation data; select per-feature thresholds to meet an agreed precision/recall trade-off.
- Use a disagreement bucket: when models differ materially, show UNKNOWN or request a human check.
- Stratify evaluation by lighting, crowding, camera angle, produce type, and occlusion.
- Run shadow mode after deployment; sample live outcomes and monitor drift. Re-evaluate after camera repositioning, display changes, or software updates.

6. SaaS architecture, rollout, decisions

Component	Attendance addition
Web application	New Attendance section, employee self-service, manager exception queue, timesheet approvals.
Identity	Tenant-aware users, roles, optional SSO, PIN/badge enrollment and account recovery.
Attendance service	Schedules, punch events, policy engine, corrections, approvals, exports.
Data store	Separate tenant-scoped attendance tables and audit events.
Integrations	HR roster import; payroll export/API; optional approved terminal hardware.
Analytics	Aggregated staffing coverage reports; camera occupancy stays anonymous and distinct.

Delivery sequence

Phase	Deliverable
A. Policy and discovery	Country/legal review, attendance rules, roles, payroll format, supported punch methods.
B. Core attendance	Roster, schedules, authenticated punches, breaks, corrections, audit log.
C. Manager workflow	Exception handling, approvals, exports, employee review.
D. Multi-model analytics	Baseline and ensemble in shadow mode for traffic/bin tasks; verify latency and quality.
E. Optional occupancy	Anonymous zone counts only; no identity matching; assess purpose and staff notice.
F. Pilot and rollout	One store, payroll reconciliation, feedback, monitoring, tenant-safe expansion.

Inputs needed to finalize the design

- Deployment country and applicable attendance/payroll rules.
- HR/payroll system, roster format, overtime/break rules, and export requirements.
- Store count, employee count, shift patterns, and shared-device/offline needs.
- Camera models/placement, edge hardware, and whether staff-only zones are covered.
- Target metrics and acceptable alert burden for each camera feature.

7. Risk controls and references

Risk	Control
Unreliable identity from video	Use credentialed clock events; never match face/track to employee attendance.
Payroll error	Employee review, manager approval, audit history, reconciliation before export.
Model disagreement	Confidence calibration; UNKNOWN state; human review for consequential cases.
Worker monitoring creep	Purpose separation, staff notice, access controls, impact review, no productivity scoring by camera.
Tenant data leak	Tenant-scoped authorization, automated access-control review, audit logs, encrypted backups.
Model drift	Versioned models/configuration, live sampling, rollback, revalidation after site changes.

Source notes

1. EDPB, Guidelines 3/2019 on processing personal data through video devices: https://www.edpb.europa.eu/documents/guideline/guidelines-32019-on-processing-personal-data-through-video-devices_en

2. UK ICO, Data protection and monitoring workers: https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/employment/monitoring-workers/data-protection-and-monitoring-workers/

3. ONVIF Profile T camera streaming: https://www.onvif.org/profiles/profile-t/

4. Ultralytics tracking documentation (one possible tracking implementation): https://docs.ultralytics.com/modes/track

5. This addendum is a product and technical plan, not jurisdiction-specific legal advice. Validate employment, payroll, privacy, CCTV, and evidence requirements locally before rollout.

End of addendum